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7.3 Rectangles, Rhombi, and Squares

Rectangles, rhombi, and squares are all parallelograms, so they inherit every parallelogram property (opposite sides parallel and congruent, opposite angles congruent, consecutive angles supplementary, diagonals bisect each other) — plus one extra rule each. A **rectangle** has four right angles, which forces its diagonals to be *congruent*. A **rhombus** has four congruent sides, which forces its diagonals to be *perpendicular* and to *bisect the vertex angles*. A **square** is both a rectangle and a rhombus, so it has all of these properties at once. Because a rhombus's diagonals are perpendicular and bisect each other, each half-diagonal pair and a side form a right triangle — the Pythagorean Theorem connects diagonal lengths to side lengths. This worksheet uses all of these relationships, often in combination.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Diagonal $AC = 22$. Find BD.

2. Perimeter 60. Each side?

3. Diagonals 10 and 24. Side length?

4. Side 7. Diagonal?

5. Diagonal $6\sqrt{2}$. Side?

6. One angle $55°$. Adjacent angle?

7. Sides 5 and 12. Diagonal?

8. Four congruent sides and four right angles. Name the figure.

9. Diagonals 8 and 6. Side?

10. Do diagonals of a rectangle always bisect its angles?

11. Perimeter 36. Diagonal?

12. Side 25, one diagonal 14. Find the other diagonal.

13. Diagonals of a rhombus are always…

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14. A rhombus has diagonals in ratio $3:4$ and area 96. Find both diagonal lengths and the side length.

15. The diagonal of a square is 4 cm longer than its side. Find the side length to the nearest tenth of a centimeter.

✓ Answer Key

1. 22
2. 15
3. 13
4. $7\sqrt{2}$
5. 6
6. $125°$
7. 13
8. Square
9. 5
10. No (only if it's a square)
11. $9\sqrt{2}$
12. 48
13. Perpendicular bisectors of each other, and they bisect the vertex angles
14. Diagonals 12 and 16; side 10
15. ≈ 9.7 cm